

Collision Lab Report

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February 23, 2026

1 Purpose

To investigate changes in momentum and kinetic energy during elastic and inelastic collisions.

2 Ground Rules

- Momentum, p , is conserved in isolated systems.
- Kinetic energy, K , is conserved in elastic collisions.
- Kinetic energy K , is not conserved in inelastic collisions.
- $p = mv$
- $K = \frac{1}{2}mv^2$

3 Data

3.1 Elastic Collisions

Table 1: Elastic Collision Data

Collision	Mass 1 (kg)	Mass 2 (kg)	Initial Vel 1 (m s ⁻¹)		Initial Vel 2 (m s ⁻¹)		Final Vel 1 (m s ⁻¹)		Final Vel 2 (m s ⁻¹)	
			x	y	x	y	x	y	x	y
Elastic 1	0.50	1.50	3.00	0.00	0.00	0.00	-1.50	0.00	1.50	0.00
Elastic 2	1.00	1.00	0.76	0.70	-0.74	0.07	-0.62	-0.13	0.64	0.90
Elastic 3	0.10	3.00	0.33	0.00	-3.00	0.01	-6.11	0.12	-2.79	0.01

3.2 Inelastic Collisions

Table 2: Inelastic Collision Data

Collision	Mass 1 (kg)	Mass 2 (kg)	Initial Vel 1 (m s ⁻¹)		Initial Vel 2 (m s ⁻¹)		Final Vel 1 (m s ⁻¹)		Final Vel 2 (m s ⁻¹)	
			x	y	x	y	x	y	x	y
Inelastic 1	0.50	1.50	2.84	0.00	0.00	0.00	0.61	0.06	0.74	-0.02
Inelastic 2	1.00	1.00	1.61	-0.17	-0.85	-0.50	0.62	-0.81	0.14	0.14
Inelastic 3	0.10	3.00	0.85	0.01	-3.00	0.01	-3.05	-0.17	-2.87	0.02

3.3 Images

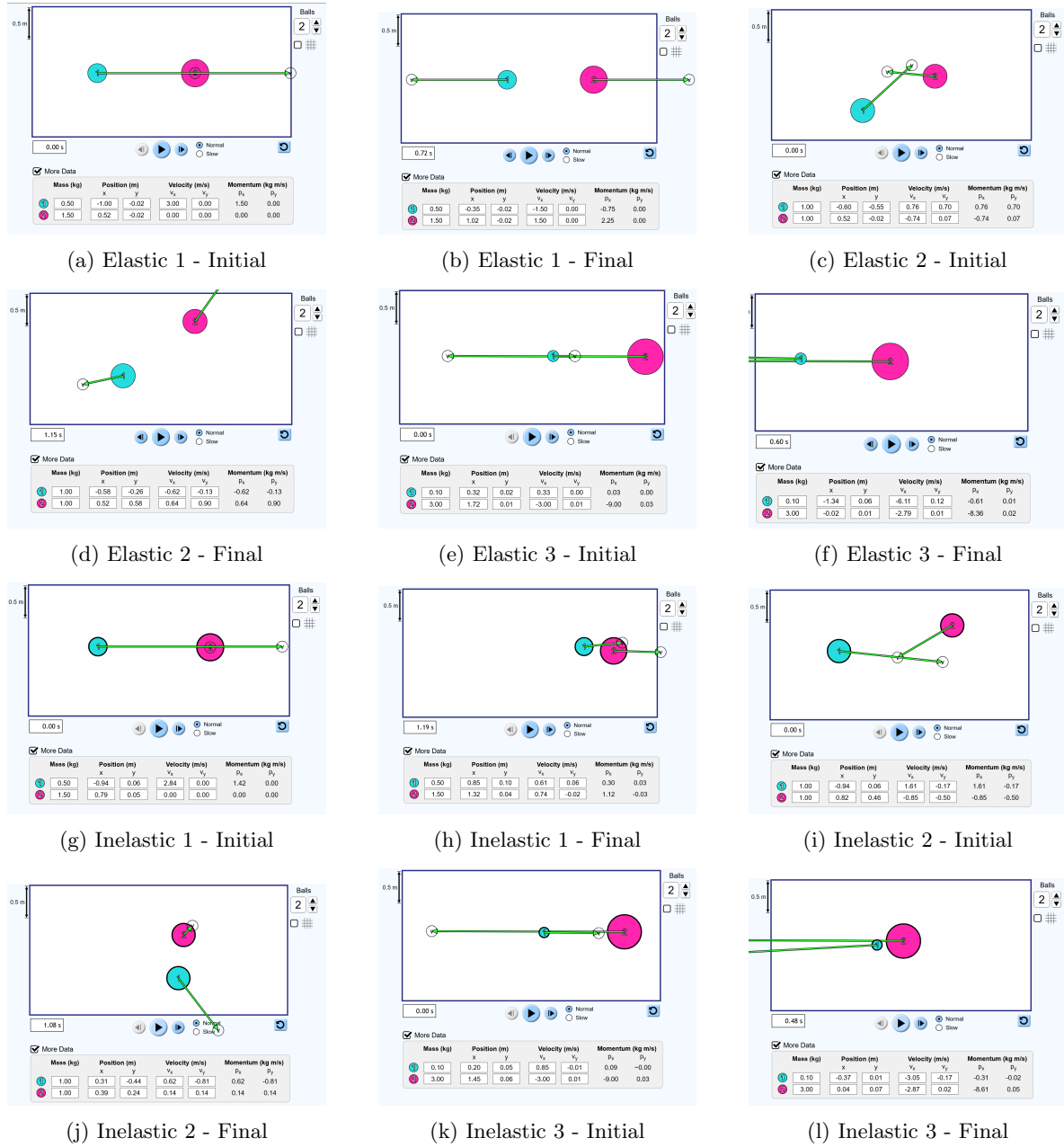


Figure 1: Collision Images

4 Analysis

4.1 Method

For each trial and each object:

$$p = mvK = \frac{1}{2}mv^2$$

System totals are vector sum of momenta and scalar sum of kinetic energies. We compare initial and final totals and report Δ (difference) and a conservation verdict (momentum: magnitude tolerance 0.02 Ns; K: percent threshold 3%).

4.2 Initial Momenta

Table 3: Initial momenta (N s)

Collision	$\mathbf{p}_{1,i,x}$	$\mathbf{p}_{1,i,y}$	$\mathbf{p}_{1,i}$	$\mathbf{p}_{2,i,x}$	$\mathbf{p}_{2,i,y}$	$\mathbf{p}_{2,i}$	$\mathbf{p}_{sys,i,x}$	$\mathbf{p}_{sys,i,y}$	$\mathbf{p}_{sys,i}$
Elastic 1	1.500	0.000	1.500	0.000	0.000	0.000	1.500	0.000	1.500
Elastic 2	0.760	0.700	1.034	-0.740	0.070	0.743	0.020	0.770	0.770
Elastic 3	0.033	0.000	0.033	-9.000	0.030	9.000	-8.967	0.030	8.967
Inelastic 1	1.420	0.000	1.420	0.000	0.000	0.000	1.420	0.000	1.420
Inelastic 2	1.610	-0.170	1.619	-0.850	-0.500	0.986	0.760	-0.670	1.013
Inelastic 3	0.085	0.001	0.085	-9.000	0.030	9.000	-8.915	0.031	8.915

4.3 Final Momenta

Table 4: Final momenta (N s)

Collision	$\mathbf{p}_{1,f,x}$	$\mathbf{p}_{1,f,y}$	$\mathbf{p}_{1,f}$	$\mathbf{p}_{2,f,x}$	$\mathbf{p}_{2,f,y}$	$\mathbf{p}_{2,f}$	$\mathbf{p}_{sys,f,x}$	$\mathbf{p}_{sys,f,y}$	$\mathbf{p}_{sys,f}$	Δp	Conserved
Elastic 1	-0.750	0.000	0.750	2.250	0.000	2.250	1.500	0.000	1.500	0.000	Yes
Elastic 2	-0.620	-0.130	0.633	0.640	0.900	1.104	0.020	0.770	0.770	0.000	Yes
Elastic 3	-0.611	0.012	0.611	-8.370	0.030	8.370	-8.981	0.042	8.981	0.018	Yes - rounding error
Inelastic 1	0.305	0.030	0.306	1.110	-0.030	1.110	1.415	0.000	1.415	-0.005	Yes - rounding error
Inelastic 2	0.620	-0.810	1.020	0.140	0.140	0.198	0.760	-0.670	1.013	0.000	Yes
Inelastic 3	-0.305	-0.017	0.305	-8.610	0.060	8.610	-8.915	0.043	8.915	0.000	Yes

4.4 Kinetic Energy

Table 5: Kinetic Energy (J)

Collision	$\mathbf{K}_{1,i}$	$\mathbf{K}_{2,i}$	$\mathbf{K}_{sys,i}$	$\mathbf{K}_{1,f}$	$\mathbf{K}_{2,f}$	$\mathbf{K}_{sys,f}$	ΔK	Conserved
Elastic 1	2.250	0.000	2.250	0.563	1.688	2.250	0.000	Yes
Elastic 2	0.534	0.276	0.810	0.201	0.610	0.810	0.000	Yes
Elastic 3	0.005	13.500	13.506	1.867	11.676	13.544	0.038	Yes - rounding error
Inelastic 1	2.016	0.000	2.016	0.094	0.411	0.505	-1.511	No
Inelastic 2	1.311	0.486	1.797	0.520	0.020	0.540	-1.257	No
Inelastic 3	0.036	13.500	13.536	0.467	12.356	12.823	-0.714	No

Note that there are rounding errors due to the low precision the PhET simulation.

4.5 Example Calculations

For the Inelastic 2 trial, we can calculate all results values from our raw data.

Momenta:

$$\begin{aligned}
m_1 &= 1.00 \text{ kg}, & m_2 &= 1.00 \text{ kg} \\
v_{1,i,x} &= 1.61 \text{ m s}^{-1}, & v_{2,i,x} &= -0.85 \text{ m s}^{-1} \\
v_{1,i,y} &= -0.17 \text{ m s}^{-1}, & v_{2,i,y} &= -0.50 \text{ m s}^{-1} \\
v_{1,f,x} &= 0.62 \text{ m s}^{-1}, & v_{2,f,x} &= 0.14 \text{ m s}^{-1} \\
v_{1,f,y} &= -0.81 \text{ m s}^{-1}, & v_{2,f,y} &= 0.14 \text{ m s}^{-1} \\
p_{1,i,x} &= m_1 v_{1,i,x} = 1.00 \text{ kg} \cdot 1.61 \text{ m s}^{-1} = 1.610 \text{ N s}, & p_{2,i,x} &= -0.850 \text{ N s} \\
p_{1,i,y} &= -0.170 \text{ N s}, & p_{2,i,y} &= -0.500 \text{ N s} \\
p_{1,f,x} &= 0.620 \text{ N s}, & p_{2,f,x} &= 0.140 \text{ N s} \\
p_{1,f,y} &= -0.810 \text{ N s}, & p_{2,f,y} &= 0.140 \text{ N s} \\
|\vec{p}_{1,i}| &= \sqrt{p_{1,i,x}^2 + p_{1,i,y}^2} = 1.619 \text{ N s}, & |\vec{p}_{2,i}| &= 0.986 \text{ N s} \\
|\vec{p}_{1,f}| &= 1.020 \text{ N s}, & |\vec{p}_{2,f}| &= 0.198 \text{ N s} \\
p_{sys,i,x} &= p_{1,i,x} + p_{2,i,x} = 0.760 \text{ N s}, & p_{sys,f,x} &= 0.760 \text{ N s} \\
p_{sys,i,y} &= -0.670 \text{ N s}, & p_{sys,f,y} &= -0.670 \text{ N s} \\
|\vec{p}_{sys,i}| &= \sqrt{p_{sys,i,x}^2 + p_{sys,i,y}^2} = 1.013 \text{ N s}, \\
|\vec{p}_{sys,f}| &= \sqrt{p_{sys,f,x}^2 + p_{sys,f,y}^2} = 1.013 \text{ N s} \\
\Delta p &= |\vec{p}_{sys,f}| - |\vec{p}_{sys,i}| = 0.000 \text{ N s} \\
&\therefore \text{Momentum is conserved.}
\end{aligned}$$

Kinetic Energy:

$$\begin{aligned}
K_{1,i} &= \frac{1}{2} m_1 v_{1,i}^2 = \frac{1}{2} m_1 (v_{1,i,x}^2 + v_{1,i,y}^2) = 1.310 \text{ J}, & K_{2,i} &= 0.486 \text{ J} \\
K_{1,f} &= 0.520 \text{ J}, & K_{2,f} &= 0.020 \text{ J} \\
K_{sys,i} &= K_{1,i} + K_{2,i} = 1.797 \text{ J}, & K_{sys,f} &= 0.540 \text{ J} \\
\Delta K &= K_{sys,f} - K_{sys,i} = -1.257 \text{ J} \\
&\therefore \text{Kinetic energy is not conserved.}
\end{aligned}$$

5 Explanation

The diagrams below illustrate the tip-to-tail addition of the momentum components for the chosen collision (Inelastic 2). In each diagram the blue vector is $\vec{p}_1$, the red vector is $\vec{p}_2$ (drawn from the tip of $\vec{p}_1$), the dotted teal line is the tip-to-tail resultant $\vec{p}_1 + \vec{p}_2$, and the dashed olive arrow is the system momentum $\vec{p}_{sys}$. Component arrows are shown as dashed lines; numeric component values are listed below in equation form using units of N s.

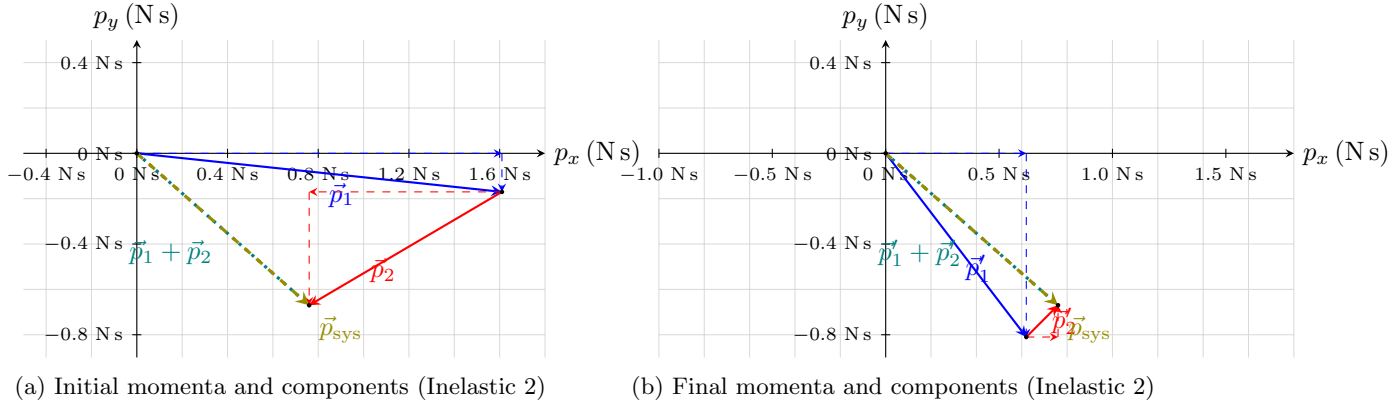


Figure 2: Tip-to-tail addition of momentum components for Inelastic 2; the tip-to-tail resultant (dotted) matches the system momentum (dashed), confirming momentum is conserved.

Initial (Inelastic 2): $p_{1,i,x} = 1.610 \text{ N s}$, $p_{1,i,y} = -0.170 \text{ N s}$,
 $p_{2,i,x} = -0.850 \text{ N s}$, $p_{2,i,y} = -0.500 \text{ N s}$,
 $p_{sys,x} = p_{1,i,x} + p_{2,i,x} = 0.760 \text{ N s}$, $p_{sys,y} = p_{1,i,y} + p_{2,i,y} = -0.670 \text{ N s}$

Final (Inelastic 2): $p_{1,f,x} = 0.620 \text{ N s}$, $p_{1,f,y} = -0.810 \text{ N s}$,
 $p_{2,f,x} = 0.140 \text{ N s}$, $p_{2,f,y} = 0.140 \text{ N s}$,
 $p_{sys,x} = p_{1,f,x} + p_{2,f,x} = 0.760 \text{ N s}$, $p_{sys,y} = p_{1,f,y} + p_{2,f,y} = -0.670 \text{ N s}$